

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: BVCA-202 — Problem Solving AI

Time: Three Hours

Maximum Marks: 70

Instructions: Answer all questions in Part A, any four questions from Part B, and any three questions from Part C.

PART — A

Note: Answer all questions. Each question carries 2 marks.

[10 × 2 = 20]

Question 1. Answer the following:

- (a) Briefly explain the Turing Test.
- (b) Discuss the Chinese Room Argument.
- (c) Describe the PEAS model on which an AI agent works.
- (d) Transform the following English sentences into logical notation:
 - (i) Not all integers are odd.
 - (ii) Some numbers are not real.
- (e) Write the following rules of inference related to the language of propositions:
 - (i) Modus Ponens
 - (ii) Hypothetical Syllogism
- (f) Which of the following is not a logic programming language?
 - (i) Prolog
 - (ii) ROOP
 - (iii) Janus
 - (iv) C
- (g) Which of the following is not a heuristic search?
 - (i) Best First Search
 - (ii) Hill Climbing
 - (iii) Depth First Search
 - (iv) A* Search
- (h) Which of the following is not a proposition?

- (i) $X = 3$
 - (ii) Moon is made up of cheese.
 - (iii) $X > 6$
 - (iv) $3 + 3 = 6$
- (i) Which of the following is false, where the underlying universe of discourse is the set of integers?
- (i) $\forall x[x < x + 1]$
 - (ii) $\exists x[x < x + 1]$
 - (iii) $\exists x[x = x + 1]$
 - (iv) $\forall x[x = x]$
- (j) Which of the following techniques is not a structured knowledge representation technique?
- (i) Semantic Nets
 - (ii) Production Rules
 - (iii) Scripts
 - (iv) Frames

PART — B

Note: Answer any four questions from the following. Each question carries 5 marks. $[4 \times 5 = 20]$

Question 2. Give examples of two agents with their PEAS representation.

Question 3. Explain the Hill Climbing Algorithm along with its types.

Question 4. If horses fly or cows eat grass, then the mosquito is the national bird. If the mosquito is the national bird, then peanut butter tastes good on hot dogs. But peanut butter tastes terrible on hot dogs. Therefore, cows don't eat grass. Prove or disprove this argument using the resolution principle.

OR

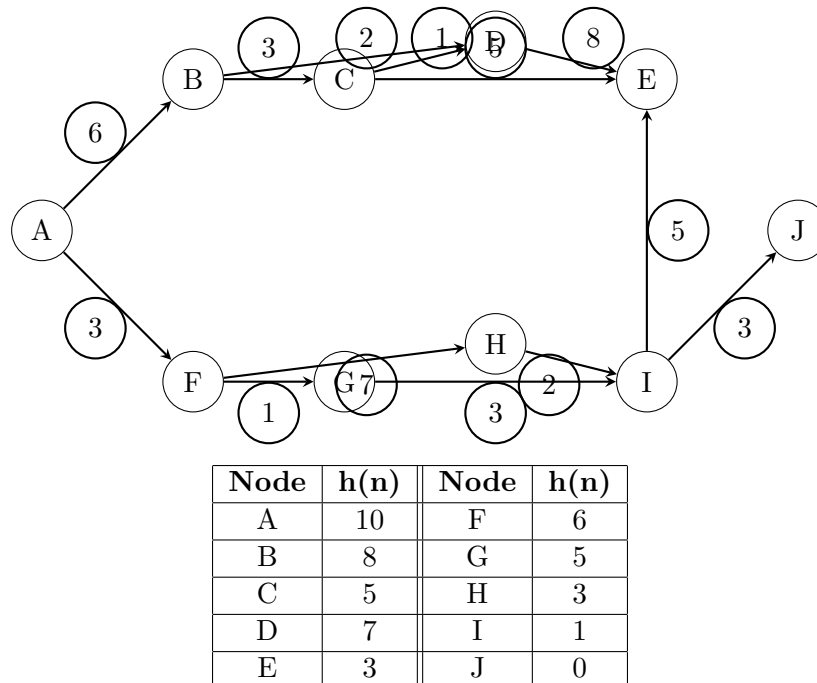
Prove or disprove the validity of the following argument with respect to the given statements:

- (a) David is a professor.
- (b) All professors are people.
- (c) John is the dean.
- (d) Deans are professors.
- (e) All professors consider the dean a friend or don't know him.
- (f) Everyone is a friend of someone.
- (g) People only criticize people that are not their friends.

(h) David criticizes John.

(i) **Conclusion:** John is not a friend of David.

Question 5. Consider the following graph. Find the most cost-effective path to reach from start state *A* to final state *J* using the *A** Algorithm:



OR

Write the pseudo-code for the Min-Max algorithm and explain its working with an example.

Question 6. Write the advantages and disadvantages of the logical representation technique of knowledge representation.

OR

Write a Prolog program to create the following knowledge base and return the answers for the queries:

(a) **Knowledge Base:**

- Lili is happy if she dances.
- Tom is hungry if he is searching for food.
- Jack and Bill are friends if both of them love to play cricket.
- Ryan will go to play if the school is closed and he is free.

(b) **Queries:**

- Is Tom a cat?
- Does Kunal love to eat pasta?
- Is Lili happy?
- Will Ryan go to play?

PART — C _____

Note: Answer any three questions from the following. Each question carries 10 marks. [3 × 10 = 30]

Question 7. Explain different types of Agents defined based on their degree of perceived intelligence and capability.

Question 8. Explain how the resolution principle is used in the Prolog language by an example.

Question 9. Write the Best First Search algorithm. What is the difference between Greedy versus A^* Best First Search Algorithm?

Question 10. What do you understand by structured knowledge representation techniques? Explain any one of these techniques.